Scheme 2

Stage 2 Abstract-Level Structured Coding Scheme

Prompt-guided abstract coding with deterministic repair and rule-based fallback

Purpose

This scheme standardizes abstract-level extraction for all eligible chronic pain reviews. It is the main corpus-wide coding layer used to describe review characteristics, classify the function of biopsychosocial language, identify the presence of biological, psychological, and social content, flag conceptual problems, and generate a provisional biopsychosocial typology for downstream synthesis.

Operational Status

- Workflow position: Stage 2 abstract coding after Stage 1 eligibility screening.
- Operational mode: structured LLM-first coding with archived JSON batch outputs, deterministic vocabulary normalization, and rule-based fallback when model output is incomplete or unavailable.
- Canonical implementation basis: actual Stage 2 output schema in `src/review_stages/04_extraction/outputs/stage2_abstract_coding.csv`.

Canonical Source Paths

- `src/protocol/codebooks/stage2_codebook.md`
- `src/review_stages/04_extraction/codebooks/stage2_codebook.csv`
- `src/bps_review/extraction/stage2.py`
- `src/bps_review/extraction/llm_stage2.py`
- `src/review_stages/04_extraction/outputs/stage2_abstract_coding.csv`
- `src/review_stages/04_extraction/outputs/stage2_llm_structured_coding.csv`
- `src/review_stages/04_extraction/outputs/llm_stage2_structured_batches.jsonl`

Carried-Through Metadata Fields

`record_id` Internal record identifier.

`database` Source database.

`pmid` / `pmcid` / `doi` External identifiers when available.

`title` / `abstract` Primary text used for coding.

year / journal / authors / country_contact_author
Descriptive metadata retained for synthesis.

publication_types
PubMed or source publication-type metadata.

objective_text Objective sentence or excerpt extracted from the abstract.

screening_status / screening_reason
Stage 1 status inherited into the Stage 2 dataset.

Coded Fields and Controlled Values

review_type `systematic review`; `meta-analysis`; `network meta-analysis`; `umbrella review`; `scoping or mapping review`; `rapid review`; `realist review`; `integrative review`; `narrative or expert review`; `other evidence synthesis`; `unclear`.

objective_category
`conceptual`; `clinical`; `methodological`; `epidemiological`; `mixed`; `unclear`.

objective_category_source
Indicates whether the objective classification came from the structured LLM, a repaired LLM batch, or the deterministic fallback.

icd11_pain_category
`chronic secondary musculoskeletal pain`; `chronic neuropathic pain`; `chronic cancer-related pain`; `chronic postsurgical or posttraumatic pain`; `chronic secondary headache or orofacial pain`; `chronic secondary visceral pain`; `chronic primary pain`; `mixed or unspecified chronic pain`; `unclear`.

musculoskeletal_flag
`yes`; `no`; `unclear`.

bps_mention_location
`title only`; `abstract only`; `title and abstract`; `unclear`.

bps_function `explanatory framework`; `intervention rationale`; `organizing principle`; `justification`; `background framing`; `conclusion`; `policy/practice implication`; `rhetorical label`; `unclear`.

bio_mentioned / psych_mentioned / social_mentioned
`yes` or `no`.

quality_assessment_reported
`yes`; `no`; `unclear`.

psychological_concepts_detected
Normalized list of psychological concepts extracted from title and abstract.

theoretical_frameworks_detected
Normalized list of named frameworks or model labels.

`conceptual_problem_flags`
: `vague_definition`; `tokenistic_bps`; `missing_social`; `missing_biology`; `mechanistic_absence`; `construct_overlap`; `parallel_listing_without_integration`; `none`.

`provisional_typology`
: `potential integrative signal`; `multifactorial signal`; `pseudo-bps or partial signal`; `rhetorical label signal`.

`stage3_candidate` `yes` or `no`.

`stage3_priority` `high`; `medium`; `low`.

`coding_rationale` One-sentence rationale supporting the coding bundle.

`coding_method` Operational provenance of the row, for example `llm_structured`, `llm_batch_fallback`, or `rule_based_fallback`.

`llm_model` Model identifier when LLM coding was used.

Prompt Rules Used in the Structured LLM Stage

- Code only from title, abstract, publication types, and journal metadata.
- Do not infer content that is absent from the record.
- Do not treat the single lexical token `biopsychosocial` as proof that biological, psychological, and social domains are all substantively present.
- Use `explanatory framework` only when the review explicitly treats BPS as a model or framework explaining pain or pain-related disability.
- Use `intervention rationale` when BPS language mainly justifies multimodal or interdisciplinary treatment.
- Use `organizing principle` when BPS language structures the scope or categories without specifying integration mechanisms.
- Use `rhetorical label` when BPS is invoked ceremonially, aspirationally, or symbolically without substantive analytic work.
- Set `stage3_candidate` to `yes` for musculoskeletal reviews and mixed or unspecified chronic pain reviews where musculoskeletal relevance cannot be ruled out.
- Use `potential integrative signal` only when all three core domains are substantively present and the abstract signals cross-domain explanation or organization beyond simple listing.
- Use `multifactorial signal` when all three domains are substantively present but mainly parallel.
- Use `pseudo-bps or partial signal` when one or more core domains are thin or absent despite BPS language.

- Use `rhetorical label signal` when the BPS label is mainly symbolic or confined to framing or conclusion.
- Return `conceptual_problem_flags` as `none` only when no inferable conceptual problem is present.

Deterministic Repair and Fallback Logic

- Missing or malformed model responses are repaired against the fixed vocabularies listed above.
- Aliases such as `scoping review` or `narrative review` are normalized to the implemented labels.
- `stage3_candidate` is forced to `yes` whenever `musculoskeletal_flag` is `yes` or `unclear`.
- Conceptual problem flags are post-derived from typology, BPS function, and missing domains. Rhetorical usage triggers `tokenistic_bps` and `vague_definition`; parallel non-mechanistic usage triggers `parallel_listing_without_integration` and `mechanistic_absence`; absent biological or social content triggers `missing_biology` or `missing_social`.
- When the LLM stage fails, the rule-based fallback still generates review type, objective category, ICD-11 category, BPS function, domain mention flags, concept detections, and provisional Stage 3 priority.

Implementation Notes and Documented Divergences

- The prose codebook lists `treatment-oriented` as an objective class, but the implemented structured prompt and output schema do not currently use that label.
- The CSV codebook historically contains `spiritual_existential_mentioned` and `lifestyle_mentioned`; those fields are not present in the current Stage 2 output file and therefore are not treated as canonical in this dossier.
- The prose codebook and the executable code both emphasize that BPS language alone is insufficient evidence of true triadic coverage.

Primary Outputs Using This Scheme

- `src/review_stages/04_extraction/outputs/stage2_abstract_coding.csv`
- `src/review_stages/04_extraction/forms/stage2_double_code_subset.csv`
- `src/review_stages/04_extraction/outputs/stage2_objective_llm_assist.csv`
- `src/review_stages/04_extraction/outputs/llm_objective_pilot.json`